



FACULTY OF MANAGEMENT

MBA Programme

Project

For

Database Management and Mining Technologies

on

Airbnb Rental Management and Data Analysis System using PostgreSQL

Submitted to:

Dr. Amish Soni

Submitted by:

Enrollment No.	Name of the Students
202400620010201	Burhanuddin Gadiwala
202400620010205	Burhanuddin Gangardiwala
202400620010233	Isha Haria
202400620010300	Vanshika Majithia
202400620010326	Sahaj Mitra
202400620010513	Raj Shah
202400620010551	Jeet Solanki

Semester IV

Specialization – BA

Class: BA – 2

Year 2025-26

DECLARATION

We, the undersigned, hereby declare that the research work titled "Airbnb Rental Management and Data Analysis System using PostgreSQL" is an original piece of work carried out by us as part of our MBA program under the guidance of Dr. Amish Soni. The information and data presented in this project have been created, collected, and analysed by us using PostgreSQL and reliable academic and technical sources. All references used in this study have been properly acknowledged wherever necessary.

We further declare that this project has not been previously submitted to any other university, institution, or organization for any academic or professional purpose. The work presented in this report is our own and reflects our understanding and practical implementation of database management and analysis using PostgreSQL. We take full responsibility for the authenticity and accuracy of the content included in this project.

Date: _____

Submitted by:

Burhanuddin Gadiwala – 202400620010201

Burhanuddin Gangardiwala - 202400620010205

Isha Haria – 202400620010233

Vanshika Majithia – 202400620010300

Sahaj Mitra - 202400620010326

Raj Shah – 202400620010513

Jeet Solanki – 202400620010551

PREFACE

This project titled "Airbnb Rental Management and Data Analysis System using PostgreSQL" has been prepared as part of the MBA program to develop practical knowledge and understanding of database management systems. The main purpose of this project is to learn how PostgreSQL can be used to create, manage, and analyse structured data efficiently in a real-world scenario such as an Airbnb rental system.

Through this project, we gained hands-on experience in designing database tables, inserting data, and performing SQL queries to retrieve useful information. The study helped us understand how data can be organized, analyzed, and used for decision-making. It also enhanced our knowledge of PostgreSQL features such as data storage, filtering, aggregation, and analysis.

This project has been a valuable learning experience, allowing us to connect theoretical concepts with practical application. We hope that this study provides useful insights into the role of PostgreSQL in managing and analysing business data effectively.

ACKNOWLEDGMENT

We would like to express our sincere gratitude to our respected faculty guide, Dr. Amish Soni, for his valuable guidance, support, and encouragement throughout the completion of this project titled "Airbnb Rental Management and Data Analysis System using PostgreSQL." His knowledge, suggestions, and continuous motivation helped us successfully complete this project and enhance our understanding of database management and analysis.

We are also thankful to the Faculty of Management and our MBA Program for providing us with the opportunity to work on this project as part of our academic curriculum. This project helped us gain practical knowledge and hands-on experience in using PostgreSQL for database creation, management, and analysis.

We would also like to thank all those who directly or indirectly supported us in completing this project successfully. Finally, we express our gratitude to our family and friends for their encouragement and support throughout this work.

EXECUTIVE SUMMARY

The project titled "Airbnb Rental Management and Data Analysis System using PostgreSQL" focuses on designing and implementing a database system to manage and analyse Airbnb rental listing data using PostgreSQL. The main objective of this study is to understand how a relational database management system can be used to store, organize, and analyze structured data efficiently for business purposes.

In this project, a database was created using PostgreSQL to store important information about Airbnb listings, including host name, property name, property type, location, price per night, rating, and availability. Various SQL queries were performed to retrieve and analyse data, such as finding average prices by location, identifying the most expensive and highest-rated properties, counting properties in different locations, and analysing availability status. These queries helped in understanding pricing trends, property distribution, and overall performance of listings.

The project demonstrates that PostgreSQL is a reliable and efficient tool for managing large volumes of structured data. It ensures data accuracy, consistency, and easy retrieval, which are essential for business analysis and decision-making. The system developed in this project can be useful for managing rental platforms like Airbnb, helping users and businesses make informed decisions based on data analysis.

Overall, this project provided practical knowledge of database creation, data management, and analysis using PostgreSQL. It also highlighted the importance of database systems in real-world applications such as rental management, e-commerce, and online platforms.

Table of Contents

DECLARATION.....	2
PREFACE.....	i
ACKNOWLEDGMENT.....	ii
EXECUTIVE SUMMARY	iii
CHAPTER 2: RESEARCH METHODOLOGY	5
2.1 RESEARCH OBJECTIVES	5
2.2 SCOPE OF THE STUDY	5
2.3 NEED OF THE STUDY	5
2.4 RESEARCH DESIGN	6
2.5 DATA COLLECTION SOURCES	6
2.6 LIMITATION OF THE STUDY	7
LITERATURE REVIEW	8
CHAPTER 3: ANALYSIS	14
1. Creating table	14
2. Inserting data	15
3. Display all Airbnb listings where the location is Goa.....	16
4. Find the average price per night for each location.....	17
5. Find the most expensive Airbnb property.	18
6. Find the total number of properties for each location.	19
7. Display all Airbnb listings where the price per night is between 4000 and 6000.....	20
8. Display the top 3 highest-rated Airbnb properties.	21
9. Count the number of locations that have a rating of 4.5 or higher.....	22
10. Find the average rating of Airbnb properties for each property type.....	23
11. Find the number of available and unavailable properties.	24
12. Find the highest-priced property in each location.....	25
13. Find locations where the average rating is greater than 4.5.....	26
14. Find the total revenue potential per location (price × number of available properties).....	27
15. Find the second highest price property.....	28
16. Find locations having more than 3 properties.	29

17. Find the average price of available properties in each location.....	30
18. Find the total number of properties for each property type.	31
19. Find locations where both available and unavailable properties exist...	32
20. Find the top-rated property in each property type.....	33
21. Find properties whose rating is above the average rating of their respective location.....	34
CHAPTER 4: KEY FINDINGS.....	35
CHAPTER 5: CONCLUSION.....	36
CHAPTER 6: BIBLIOGRAPHY	37

CHAPTER 1: INTRODUCTION

PostgreSQL

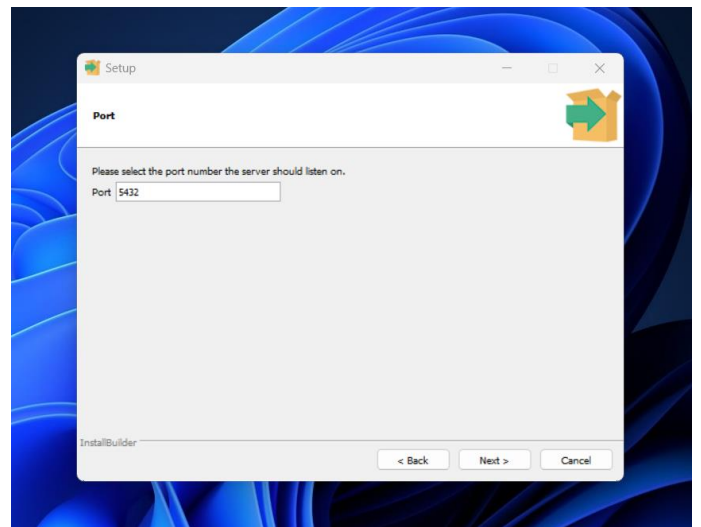
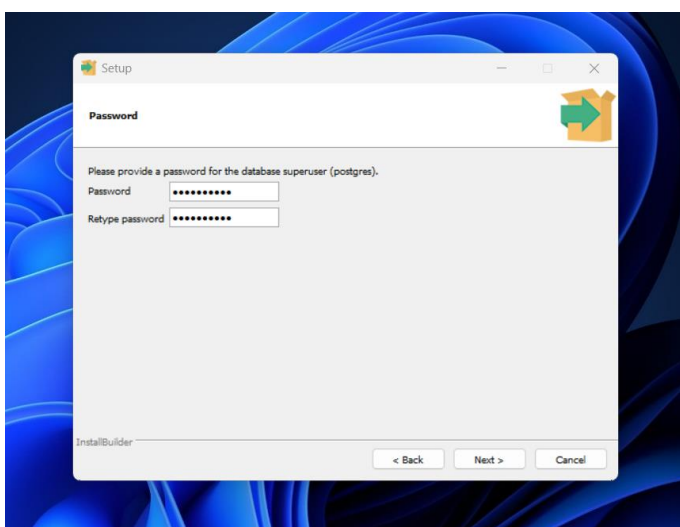
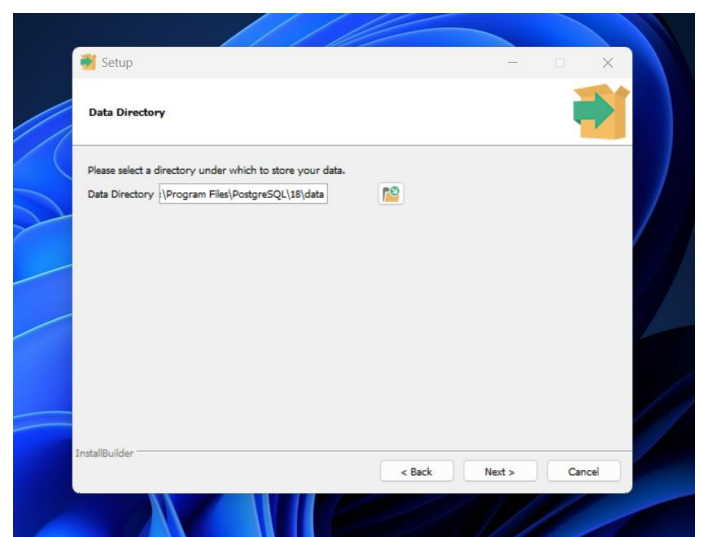
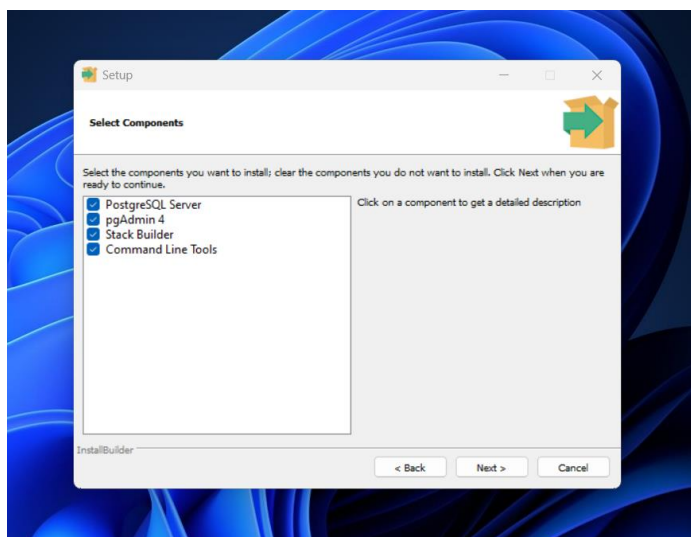
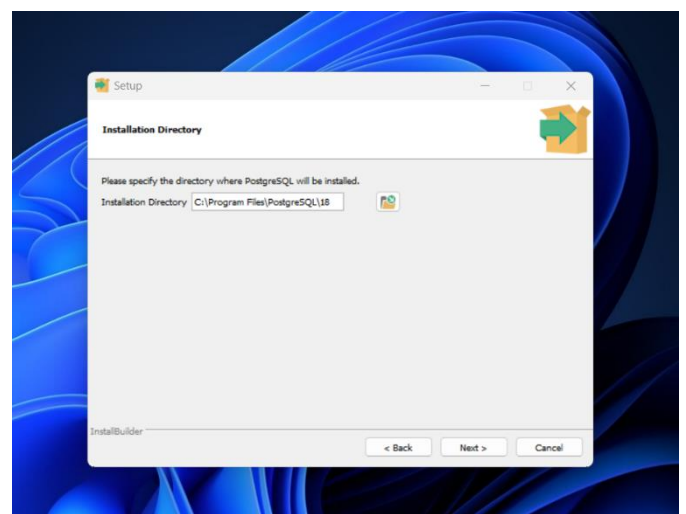
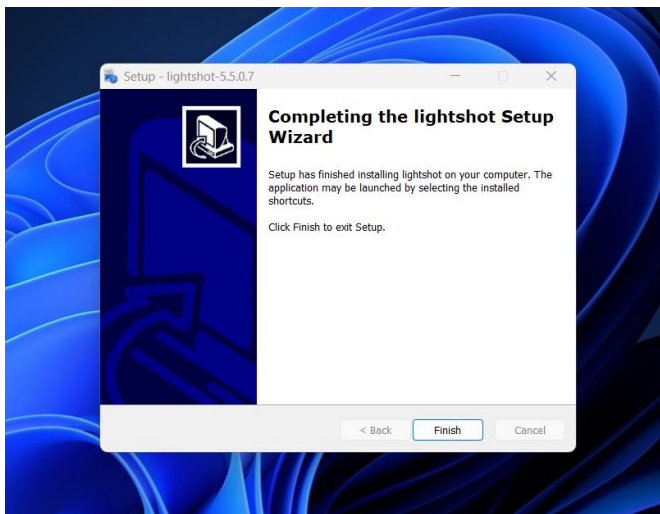
PostgreSQL is a powerful, open-source relational database management system (RDBMS) used to store, organize, and manage structured data efficiently. It supports the Structured Query Language (SQL), which allows users to perform operations such as creating databases and tables, inserting data, retrieving information, updating records, and deleting data. PostgreSQL is known for its high reliability, data integrity, scalability, and strong performance.

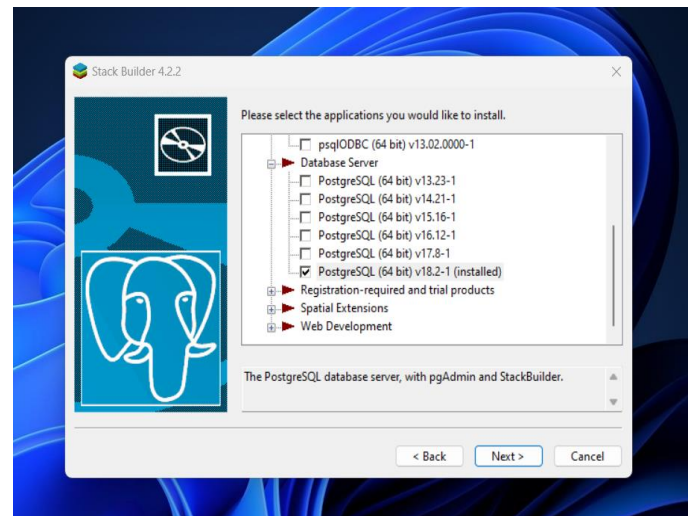
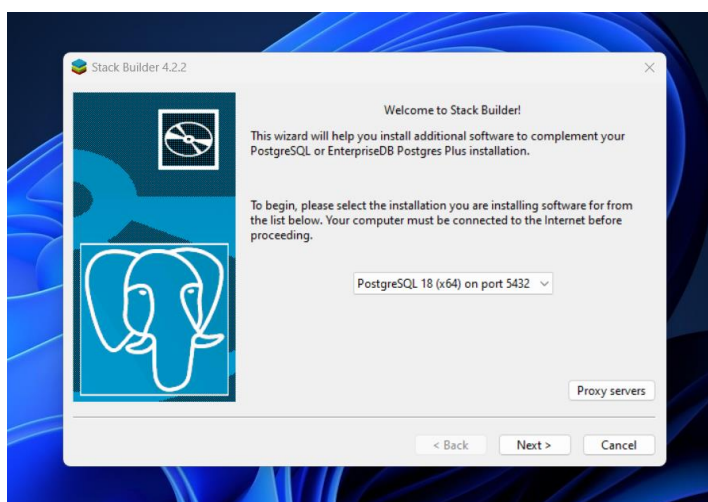
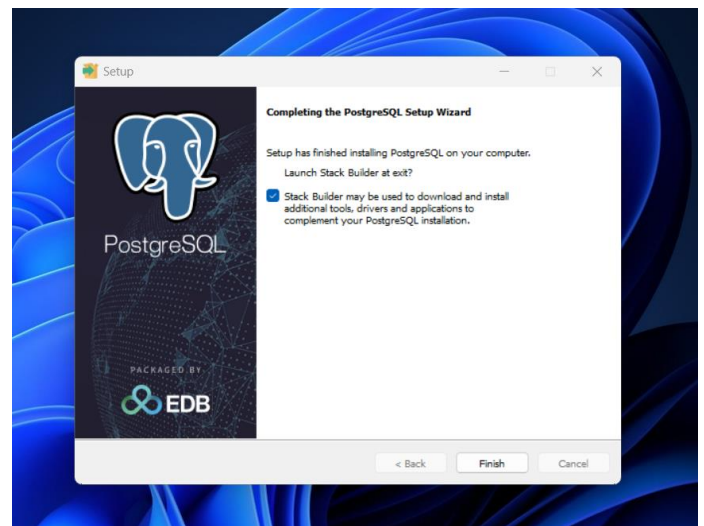
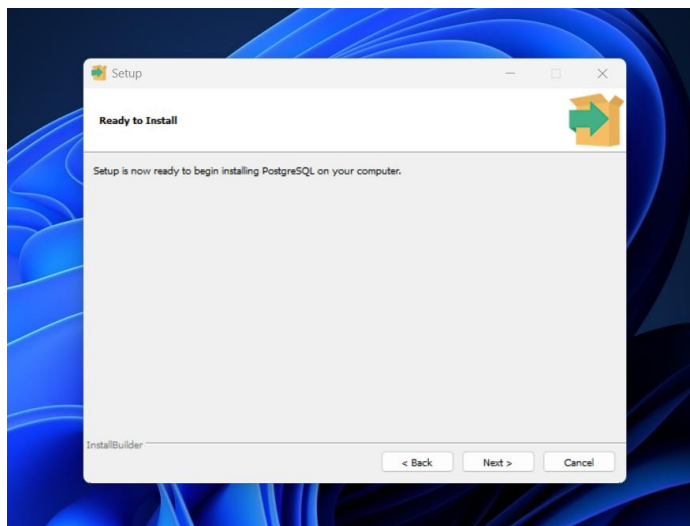
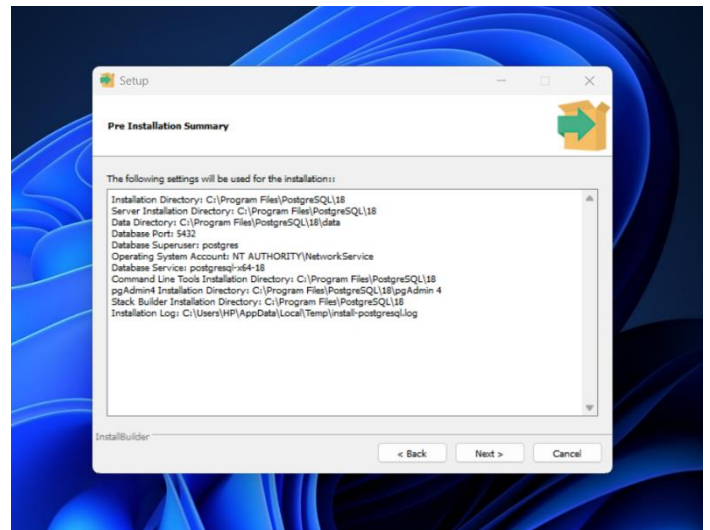
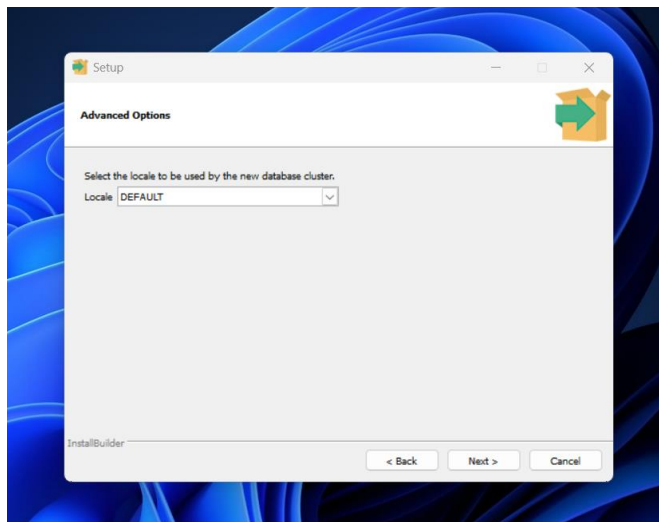
One of the key features of PostgreSQL is its ability to handle large volumes of data while maintaining accuracy and consistency. It supports advanced features such as constraints, indexing, views, and data security, which help in efficient data management and faster query execution. PostgreSQL also ensures data consistency through ACID (Atomicity, Consistency, Isolation, Durability) properties, making it suitable for real-world business applications.

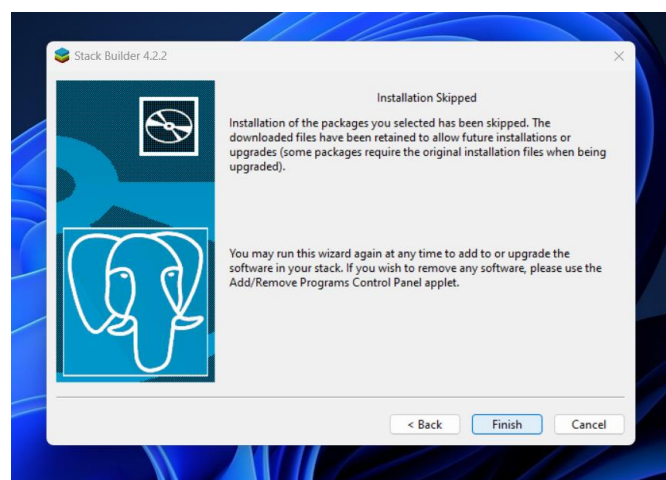
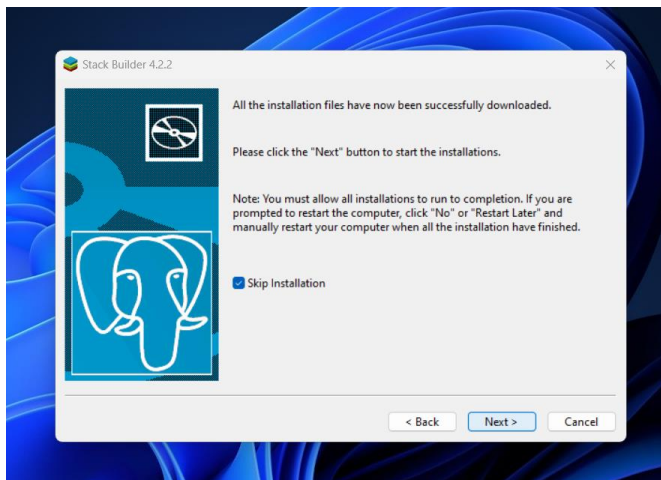
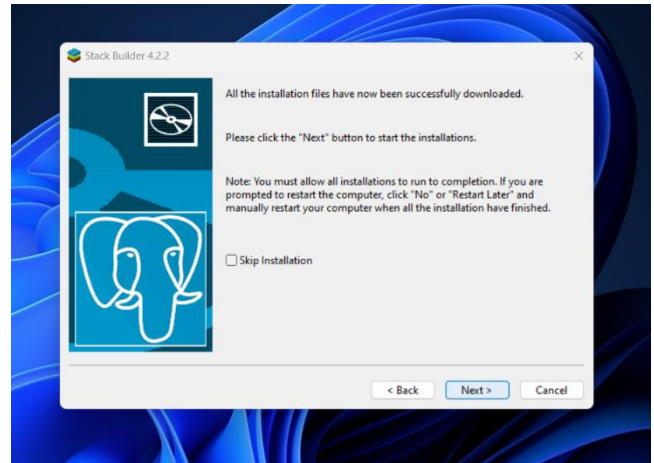
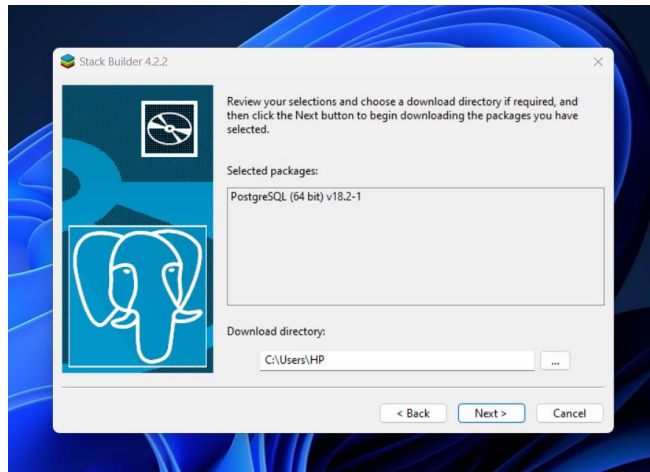
PostgreSQL is widely used in various industries such as banking, e-commerce, education, and online platforms for managing and analyzing data. It is compatible with multiple operating systems including Windows, Linux, and macOS, and is freely available for use.

In this project, PostgreSQL is used to create and manage the Airbnb rental database. It helps in storing property details, performing SQL queries, retrieving and analyzing data, and generating meaningful insights to support business decision-making.

DOWNLADING PROCESS:







CHAPTER 2: RESEARCH METHODOLOGY

2.1 RESEARCH OBJECTIVES

- To design and implement an Airbnb database using PostgreSQL.
- To retrieve and analyse Airbnb property data based on location, price, and property type.
- To identify top-rated and highest-priced properties for business insights.
- To analyze property distribution and availability across different locations.

2.2 SCOPE OF THE STUDY

This study focuses on the use of PostgreSQL for managing and analyzing Airbnb rental listing data. The scope of the study includes:

- Design and implementation of a relational database for storing Airbnb property information using PostgreSQL.
- Analysis of structured queries to retrieve, filter, and manage property data efficiently.
- Identification and evaluation of top-rated and highest-priced properties for business insights.
- Understanding the distribution of properties based on location, property type, and availability.
- Application of PostgreSQL in real-world rental management to support data analysis and decision-making.

2.3 NEED OF THE STUDY

With the rapid growth of online rental platforms like Airbnb, there is a need for an efficient database management system to store, manage, and analyze large volumes of property listing data. PostgreSQL provides a reliable and scalable solution for handling such business data effectively. This study is essential to:

- Understand the role of PostgreSQL in managing and organizing Airbnb property data efficiently.
- Help in retrieving and analyzing property information based on location, price, rating, and availability.
- Identify top-performing properties to support better business decision-making.
- Provide insights into how database systems improve data accuracy, consistency, and accessibility.
- Develop practical knowledge of PostgreSQL for real-world database management and analysis applications.

2.4 RESEARCH DESIGN

The research follows a practical approach using PostgreSQL to design and analyse an Airbnb rental database. Primary data is collected through creating tables, inserting records, and executing SQL queries in PostgreSQL. Secondary data is gathered from academic sources and technical documentation. The study focuses on data storage, retrieval, filtering, and analysis to understand how PostgreSQL supports efficient database management and business decision-making.

2.5 DATA COLLECTION SOURCES

- **Primary Data:** Database creation, data insertion, and experimentation using structured SQL queries in PostgreSQL.
- **Secondary Data:** PostgreSQL official documentation, academic books, online references, and database management study materials.

2.6 LIMITATION OF THE STUDY

- The study uses a limited dataset, which may not fully represent real-world Airbnb data.
- The analysis is performed only using PostgreSQL and does not compare with other database systems.
- The data is manually created, so it may not reflect real-time changes or actual business conditions.
- The study focuses on basic database operations and analysis, and does not include advanced features such as real-time integration or large-scale database performance testing.

LITERATURE REVIEW

1. **Cheng & Foley (2019)** This study explores how Airbnb governs hosts through algorithmic management mechanisms. Using thematic analysis of Airbnb host community discussions, the authors show that algorithms increasingly perform traditional managerial functions such as monitoring, evaluation, and control. Hosts experience both benefits (efficiency, market visibility) and challenges (uncertainty, perceived loss of autonomy). The findings highlight how algorithmic systems reshape work practices and power dynamics within platform economies. The paper contributes to understanding digital labor control in the sharing economy.
2. **Dolnicar (2018)** This book provides a comprehensive examination of peer-to-peer accommodation platforms, particularly Airbnb. It explains the unique features of platform-based hospitality models, including trust systems, network effects, and value co-creation. The work discusses economic, social, and regulatory implications such as market disruption, discrimination debates, and housing pressures. It also explores emerging themes like sustainability, accessibility, and disaster response. The book establishes a future research agenda for platform-mediated tourism.
3. **Leoni & Parker (2019)** This paper investigates governance and management control practices within Airbnb. Using a netnographic approach, the authors reveal that Airbnb relies heavily on formal bureaucratic controls despite its image as a decentralized platform. Accounting systems, surveillance mechanisms, and performance metrics are central to maintaining platform authority. Hosts' activities become visible and governable through digital monitoring. The study demonstrates how traditional control logics persist within innovative digital business models.
4. **Mody & Gomez (2018)** This article analyzes Airbnb's evolving competitive relationship with the hotel sector. The authors describe stages of industry response: denial, resistance, and strategic adaptation. Empirical evidence suggests Airbnb supply growth exerts measurable pressure on hotel performance metrics such as RevPAR. The paper highlights how hotels respond through partnerships, acquisitions, and experience-focused differentiation strategies. It frames Airbnb as both a disruptor and catalyst for industry innovation.

5. **Śpiewak-Szyjka & Oleńczuk-Paszel (2024)** This study examines operational and strategic challenges in managing short-term rental properties. Based on literature review and survey research, it identifies seasonality, maintenance costs, and service quality management as key difficulties. Digital marketing tools and booking platforms are shown to be critical for customer acquisition and efficiency optimization. The research emphasizes the professionalization of property management practices. It contributes contemporary empirical insights into short-term rental operations.
6. **Bosma & van Doorn (2024)** This paper conceptualizes the professionalization of Airbnb hosting through rent gap theory. The authors argue that Airbnb actively drives platform gentrification by privileging large-scale, professional hosts. The study distinguishes between asset-based and labor-based professionalization strategies. Evidence from Berlin illustrates uneven opportunities and growing inequalities among hosts. The work links platform dynamics with broader debates on commercialization and urban change.
7. **Gong (2023)** This dissertation explores the technological landscape of professional short-term rental (STR) property management. Using an exploratory qualitative approach, the study combines secondary desk research with in-depth interviews of STR managers and technology providers. Findings reveal strong adoption of core technologies such as Property Management Systems (PMS), channel managers, keyless entry systems, and dynamic pricing tools, with PMS identified as the central operational technology. The research highlights growing demand for integrated “all-in-one” systems to improve efficiency and guest experience. The study concludes that technology has become a critical competitive asset in professional STR management.
8. **Ndaguba & Van Zyl (2024)** This study investigates how locational and seasonal factors influence short-term rental performance using advanced spatial modeling and machine learning techniques. Analyzing STR data from Margaret River, Western Australia, the authors find that proximity to tourist attractions significantly improves ADR and occupancy rates. Seasonal variations produce predictable fluctuations, emphasizing the importance of dynamic pricing strategies. The integration of clustering algorithms (DBSCAN) and predictive models enhances understanding of market behavior. The research contributes a data-driven framework for optimizing STR revenue and operational decisions.

- 9. Avci (2020)** This thesis presents a financial feasibility study for a start-up STR business operating through Airbnb in Helsinki, Finland. The study develops an Excel-based financial planning model incorporating budgeting, cost structures, revenue assumptions, and profitability analysis. Key financial tools include cost-volume-profit analysis, contribution margin, and break-even calculations. The findings stress the strategic importance of pricing and occupancy management for revenue optimization. The research provides a structured roadmap for financial decision-making in STR ventures.
- 10. Sainaghi & d'Angella(2021)** This study explores how marketing capabilities contribute to value creation in the short-term rental sector. Based on a mixed-method approach involving quantitative data and 55 interviews, the authors identify significant differences between individual and professional hosts. Professional hosts outperform due to stronger capabilities in data analysis, dynamic pricing, customer relationship management, and revenue management practices. The findings show that revenue management software, benchmarking skills, and demand forecasting play a critical role in performance. The study explains not only the existence of performance gaps but also the underlying managerial reasons driving these differences.
- 11. Wang et al. (2024)** This paper examines methodological challenges associated with measuring short-term rental activity using open Airbnb data. The authors demonstrate that platform data limitations, including incomplete availability and booking information, complicate accurate occupancy and revenue estimation. By developing property-level income estimates and a neighbourhood vulnerability index, the study identifies areas at risk of housing stock shifts from long-term to short-term rentals. Despite data constraints, relative comparisons prove useful for policy analysis. The research highlights the importance of transparent and accessible datasets for regulatory decision-making.
- 12. Obrč & Kerbler (2021)** This study investigates Airbnb's influence on Ljubljana's long-term rental market. Using survey data and market analysis, the authors confirm that short-term rentals reduce long-term housing supply. A substantial proportion of Airbnb activity occurs within the grey economy, intensifying regulatory concerns. Financial incentives, particularly higher profitability, drive property owners to shift toward short-term rentals. The research also acknowledges potential positive urban effects while emphasizing the

need for targeted policy measures. The findings contribute empirical evidence to debates on housing affordability and STR regulation.

13. Hübscher & Feiler (2025) This study explores Fairbnb.coop as an alternative platform model within the short-term rental market. Based on qualitative interviews with stakeholders, the authors identify Fairbnb's emphasis on transparency, regulatory compliance, and community reinvestment. While the cooperative attracts socially conscious hosts and guests, challenges include limited popularity, usability constraints, and operational barriers. The study also raises concerns regarding potential greenwashing tendencies. Despite limitations, Fairbnb is positioned as a promising bottom-up self-regulatory innovation. The research contributes to discussions on platform sustainability and fairness.

14. Swor (2019) This paper analyzes the evolving legal challenges created by Airbnb and similar short-term rental platforms. The study reviews historical developments in rental markets, identifies regulatory conflicts, and evaluates existing city ordinances. Key legal concerns include zoning compliance, property registration, taxation, contractual obligations, and enforceability. The author proposes a model ordinance emphasizing health and safety standards, permitting systems, taxation frameworks, and regulatory clarity. The research contributes a practical policy framework for municipalities managing STR markets.

15. Shrestha, Devkota & Sapkota(2025) This project presents the design and development of a House Rental Management System aimed at improving rental transactions. The system addresses common inefficiencies such as fragmented listings, communication barriers, and manual record management. Using structured development methodologies, the platform integrates landlord, tenant, and admin modules. Core functionalities include property listing management, search filters, payment tracking, and database design. The project emphasizes usability, transparency, and operational efficiency within rental ecosystems.

16. Li & Biljecki (2019) This study applies machine learning and GIS techniques to examine Airbnb's influence on housing prices in Beijing. The authors develop predictive models linking Airbnb activity levels with neighborhood price sensitivity. Findings indicate heterogeneous impacts, with some neighborhoods highly sensitive while others remain

price robust. The research introduces a data-driven regulatory approach tailored to spatial variations. Policy recommendations emphasize differentiated regulation strategies based on localized market effects.

17. Camatti et al. (2024) This paper compares artificial intelligence techniques with traditional statistical models in predicting Airbnb prices. Using Dutch market data, the study incorporates financial history variables, enhancing predictive robustness. Results show that AI-based models outperform linear methods in accuracy and generalization performance. However, traditional models remain valuable for identifying predictor significance. The study highlights the benefits of hybrid modeling approaches for revenue management and pricing optimization.

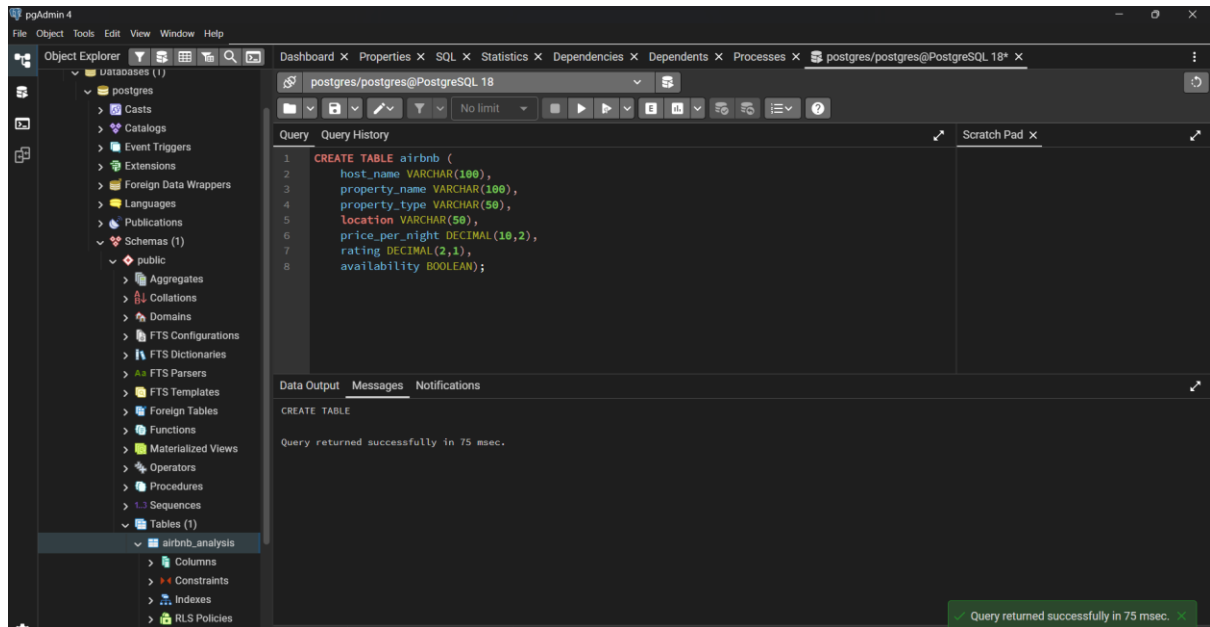
18. Deboosere et al. (2019) This study applies a multilevel hedonic regression model to analyze Airbnb listing prices and revenue in New York City. Using web-scraped transaction data, the authors find that locational factors—particularly transit accessibility to jobs—strongly influence both nightly prices and monthly revenue. The research reveals that professional hosts capture a disproportionate share of revenue, indicating increasing market professionalization. Findings suggest that converting long-term housing into short-term rentals yields significant financial premiums. The study contributes empirical evidence linking spatial economics with STR performance dynamics.

19. Sims, Ameen & Bauer This research explores pricing decision-making within Airbnb through qualitative interviews with industry professionals and hosts. The study confirms that revenue management and benchmarking principles from the hotel industry are transferable to STR markets. Findings indicate that hosts employing dynamic pricing strategies achieve improved booking performance. Differences emerge between professional and non-professional hosts in terms of pricing knowledge, motivation, and market research practices. The study highlights benchmarking as a critical capability for optimizing rates and occupancy. The research contributes managerial insights into STR pricing behavior.

- 20. Horton (2016)** This paper examines Airbnb's influence on Toronto's rental housing market, focusing on vacancy rates and affordability. Using CMHC housing data and Airbnb site-scraped listings, the study evaluates whether STR growth reduces long-term rental supply. Findings suggest that Airbnb contributes to declining vacancy rates, thereby exerting upward pressure on rental prices. The study situates Airbnb within broader structural supply constraints in Toronto's housing market. The research emphasizes the link between vacancy dynamics and rental price adjustments. The paper contributes policy-relevant empirical analysis of STR externalities.
- 21. (Filieri, 2023)** — Filieri et al. analyze large-scale Airbnb listing and booking data to quantify COVID-19's impact on booking patterns and geographic shifts in demand (urban → rural). Their work uses big-data aggregation and time-series comparisons to show structural shifts that any rental-management system must capture for accurate forecasting and pricing. For a PostgreSQL-based analytics stack the study highlights the need for scalable ingestion, normalized time-series tables, and geospatial joins (city/region) — tasks well supported by Postgres + PostGIS.

CHAPTER 3: ANALYSIS

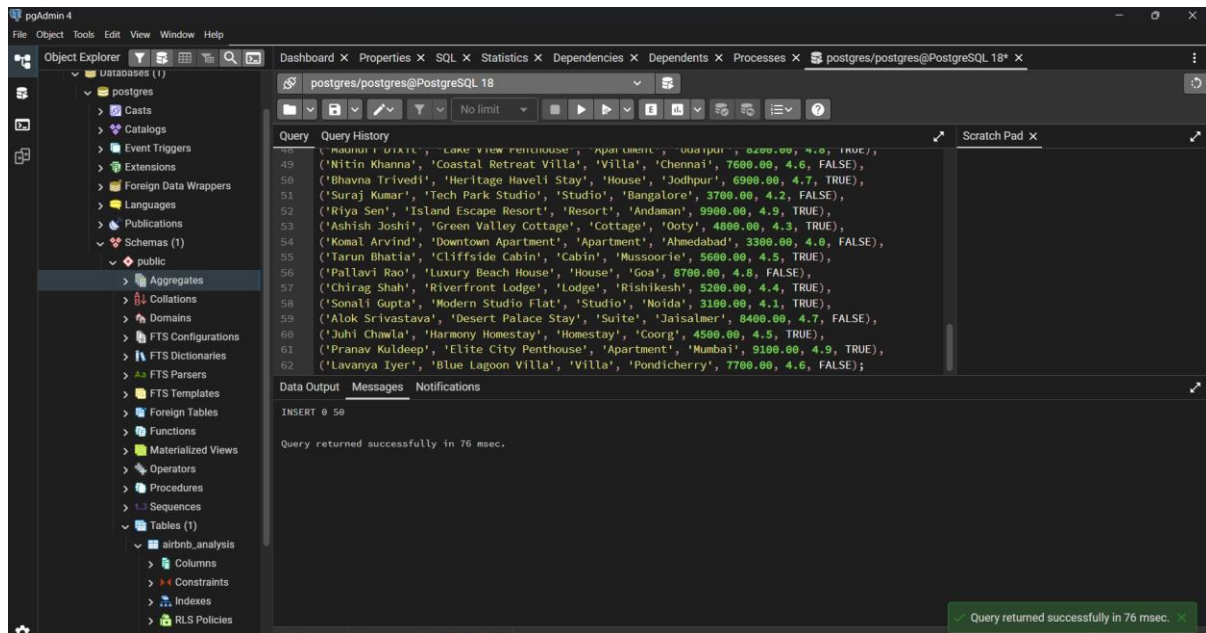
1. Creating table



```
CREATE TABLE airbnb (  
  host_name VARCHAR(100),  
  property_name VARCHAR(100),  
  property_type VARCHAR(50),  
  location VARCHAR(50),  
  price_per_night DECIMAL(10,2),  
  rating DECIMAL(2,1),  
  availability BOOLEAN);
```

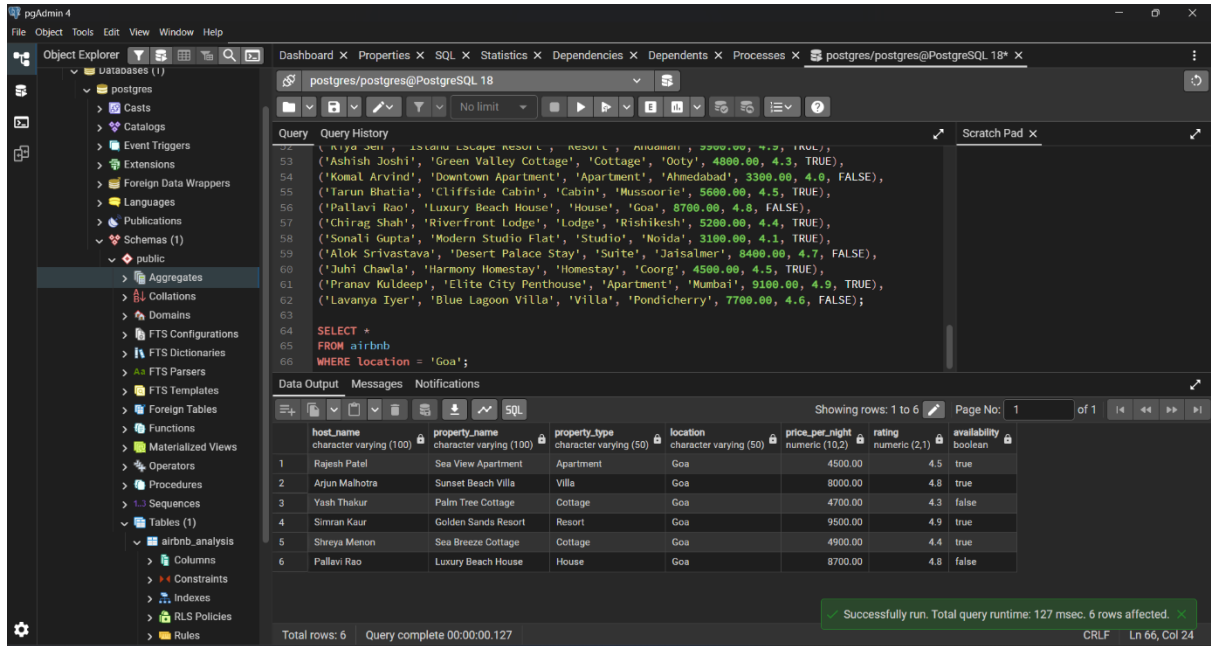
With this query, we created a table of Airbnb listings that stores information about hosts and their properties, including property name, property type, location, price per night, rating, and availability status.

2. Inserting data



The airbnb table is successfully populated with seven columns: Host_Name, Property_Name, Property_Type, Location, Price_Per_Night, Rating, and Availability. It stores detailed information about various Airbnb listings, including the host, type of property, location, nightly price, customer rating, and whether the property is available for booking.

3. Display all Airbnb listings where the location is Goa.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'public' schema and the 'airbnb_analysis' table. The main pane shows a SQL query that filters listings by location 'Goa'. The query results are displayed in a table with 6 rows. The status bar at the bottom indicates 'Successfully run. Total query runtime: 127 msec. 6 rows affected.'

```
SELECT *
FROM airbnb
WHERE location = 'Goa';
```

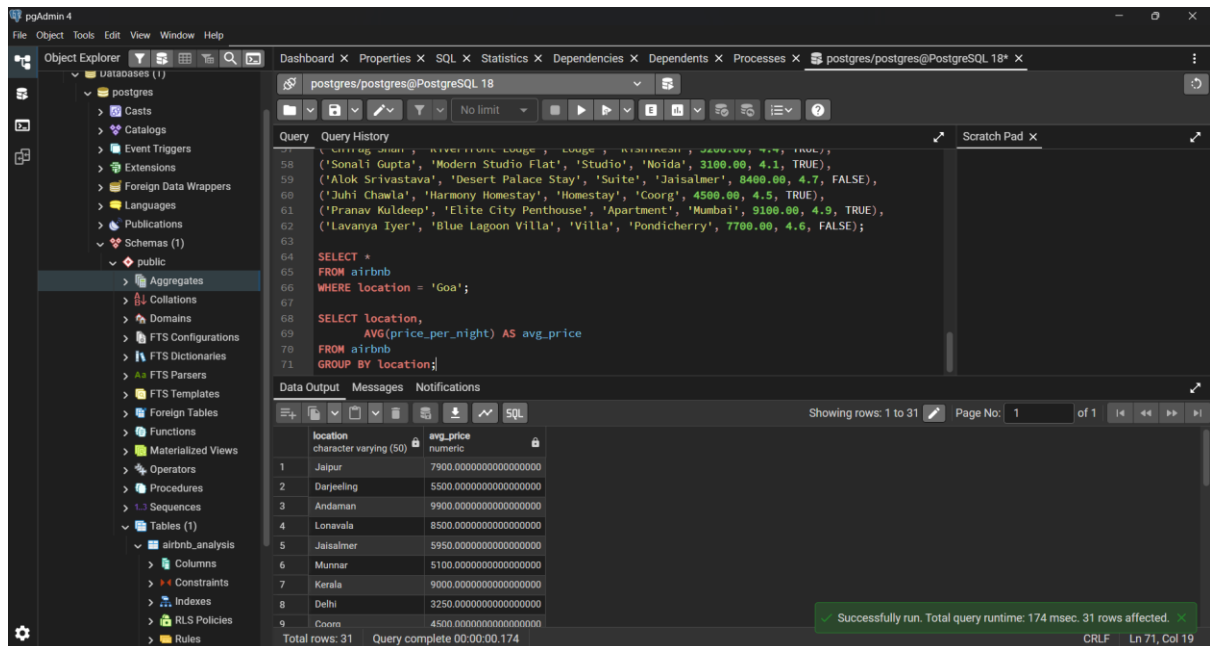
host_name	property_name	property_type	location	price_per_night	rating	availability
Rajesh Patel	Sea View Apartment	Apartment	Goa	4500.00	4.5	true
Arjun Malhotra	Sunset Beach Villa	Villa	Goa	8000.00	4.8	true
Yash Thakur	Palm Tree Cottage	Cottage	Goa	4700.00	4.3	false
Simran Kaur	Golden Sands Resort	Resort	Goa	9500.00	4.9	true
Shreya Menon	Sea Breeze Cottage	Cottage	Goa	4900.00	4.4	true

select*from airbnb

where location = 'Goa';

This query retrieves all records from the airbnb table where the location is Goa. It shows details such as the host name, property name, type of property, price per night, customer rating, and availability status. This helps identify all Airbnb listings available in Goa for analysis and comparison.

4. Find the average price per night for each location.

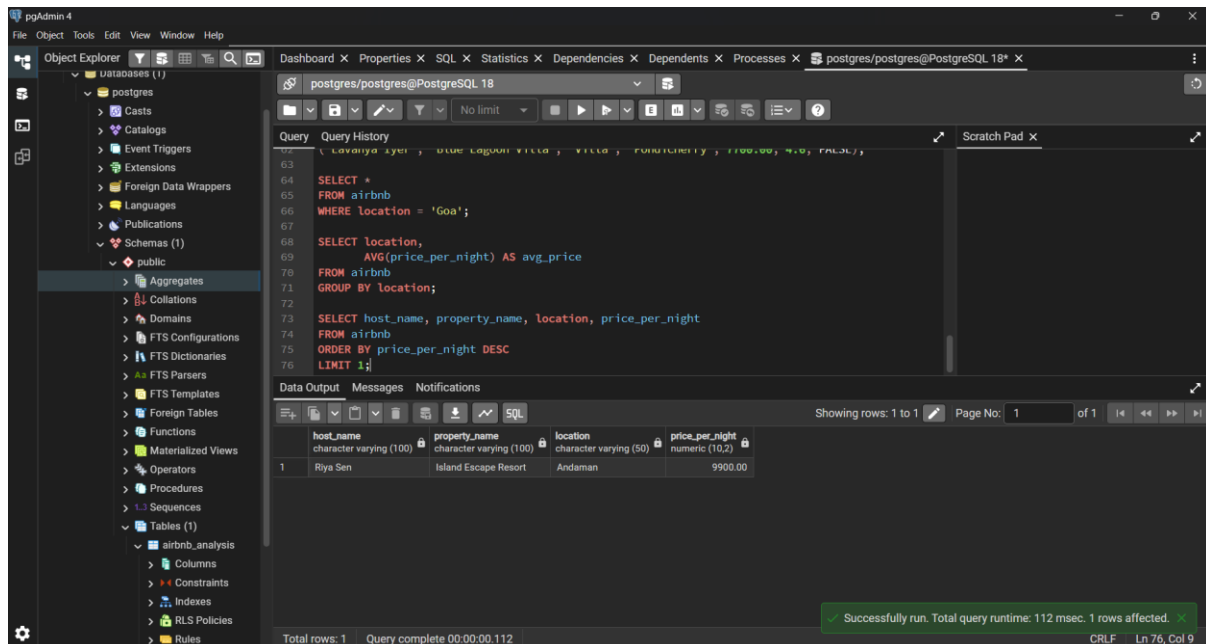


```
select location,  
avg(price_per_night) as avg_price  
FROM airbnb  
group by location;
```

This query calculates the average price per night of Airbnb listings for each location using the AVG() aggregate function. The GROUP BY location clause groups all records based on each unique location, and then the average price is calculated for each group separately.

This helps in understanding the pricing trends across different cities and allows comparison of average accommodation costs between locations. It is useful for identifying expensive and affordable destinations, analyzing market trends, and supporting decision-making for customers, hosts, or business analysis.

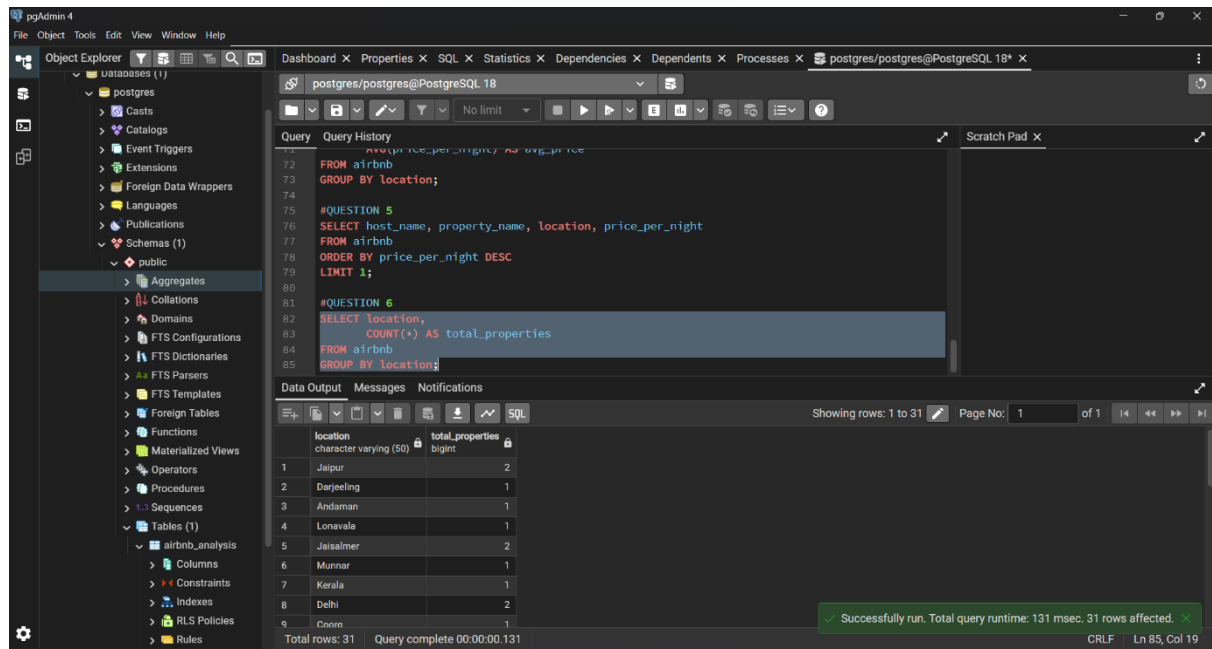
5. Find the most expensive Airbnb property.



```
select host_name, property_name, location, price_per_night  
from airbnb  
order by price_per_night desc  
limit 1;
```

This query retrieves the Airbnb property with the highest price per night. The ORDER BY price_per_night DESC clause sorts all properties from highest to lowest price, and LIMIT 1 selects only the top record. This helps identify the most expensive listing in the dataset, which is useful for price comparison, market analysis, and identifying premium properties.

6. Find the total number of properties for each location.

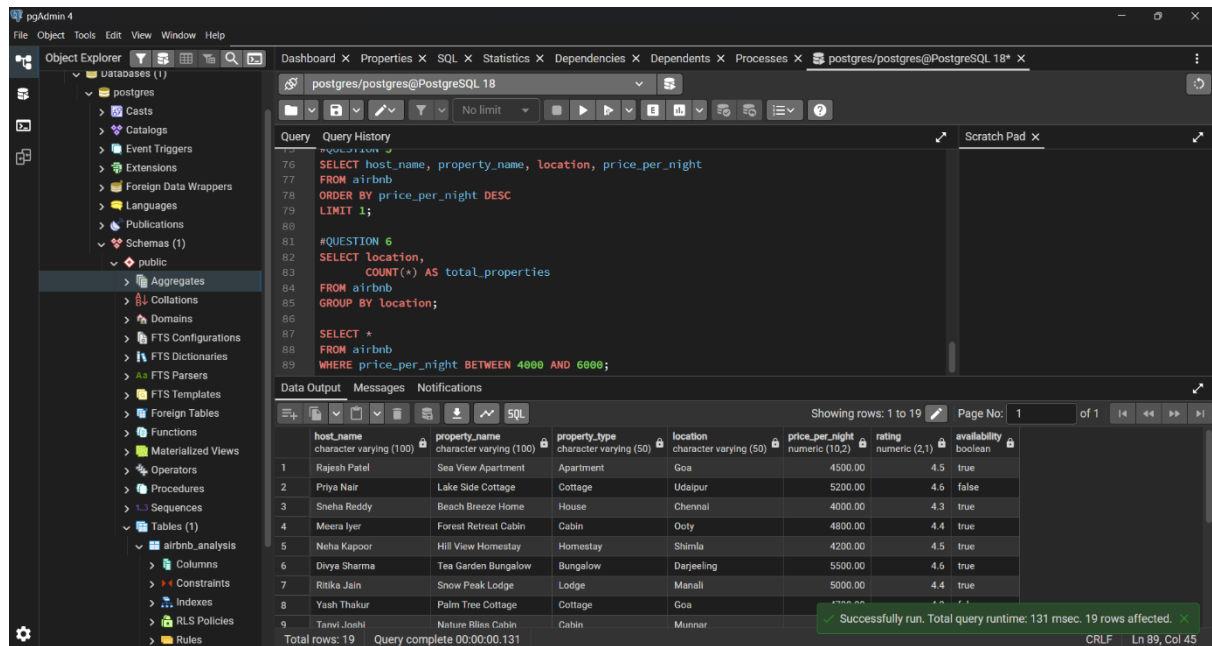


```
select location,
count(*) as total_properties
from airbnb
group by location;
```

This query counts the total number of Airbnb properties available in each location using the COUNT() function. The GROUP BY location clause groups all records based on each location, and then the total number of properties is calculated for each group.

This helps in understanding which cities have more Airbnb listings and allows comparison of property availability across different locations. It is useful for market analysis, identifying popular destinations, and evaluating supply in each area.

7. Display all Airbnb listings where the price per night is between 4000 and 6000.

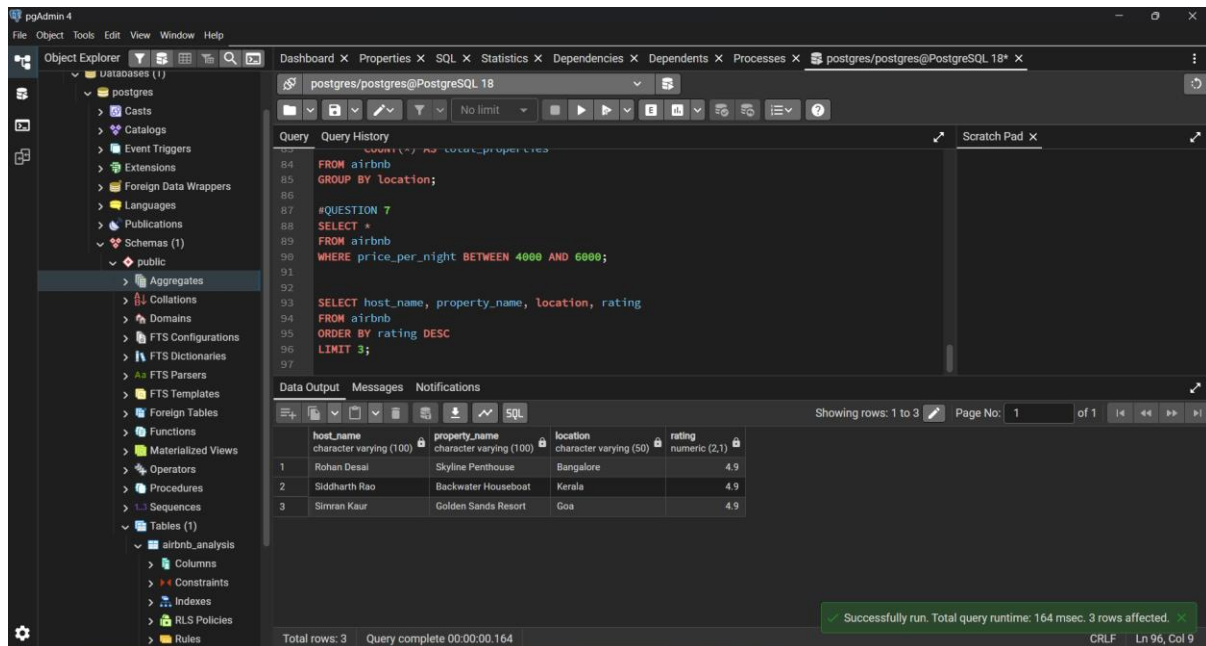


**select * from airbnb
where price_per_night between 4000 and 6000;**

This query retrieves all Airbnb listings whose price per night falls between 4000 and 6000 using the BETWEEN operator. It filters the records to show only those properties within this price range, including both 4000 and 6000.

This helps identify mid-range priced properties, making it useful for customers looking for affordable yet quality accommodations and for analyzing properties within a specific price range.

8. Display the top 3 highest-rated Airbnb properties.

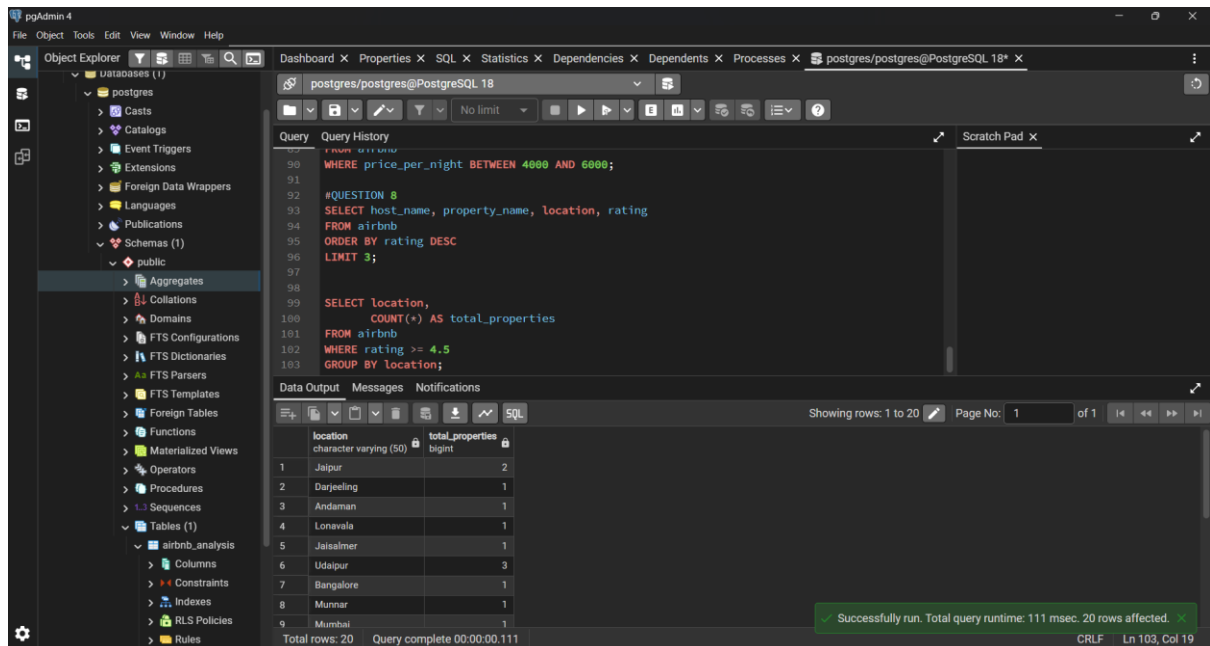


```
select host_name, property_name, location, rating
from airbnb
order by rating desc
limit 1;
```

This query retrieves the top three Airbnb properties with the highest customer ratings. The ORDER BY rating DESC clause sorts all properties from highest to lowest rating, and the LIMIT 3 clause selects only the top three records.

This helps identify the best-rated properties based on customer feedback, which is useful for customers seeking highly rated accommodations and for analyzing top-performing listings.

9. Count the number of locations that have a rating of 4.5 or higher.

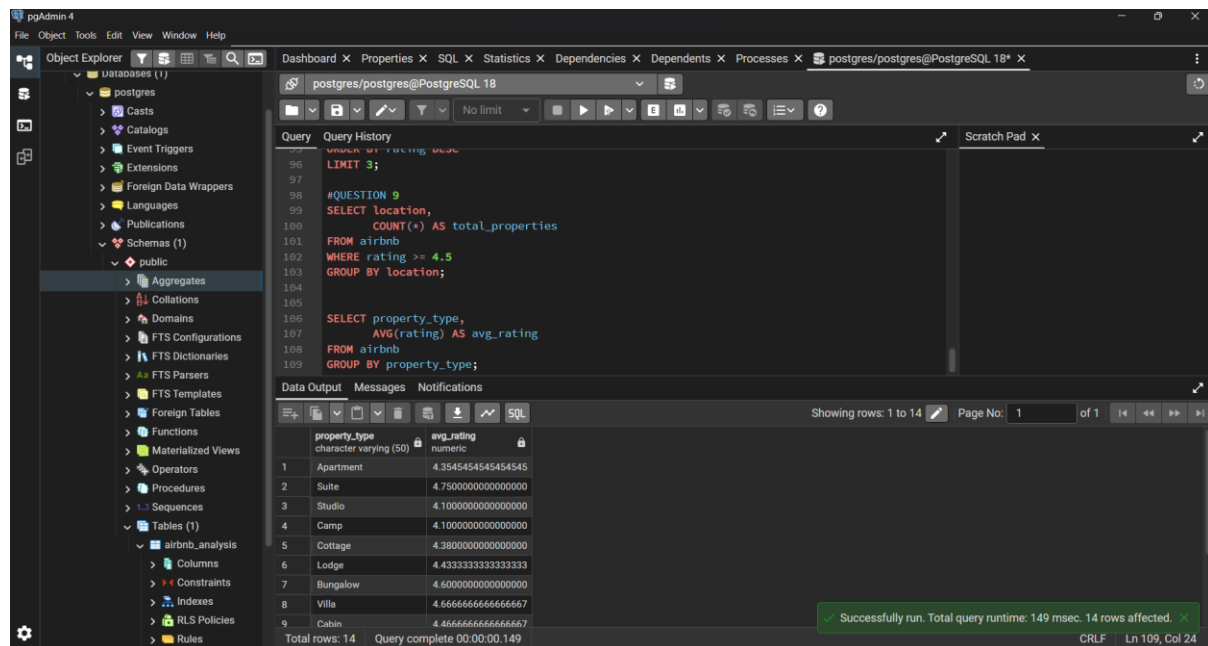


**select location,
count(*) as total_properties
from airbnb
where rating >= 4.5
group by location;**

This query counts the number of Airbnb properties in each location that have a rating of 4.5 or higher. The WHERE rating >= 4.5 clause filters the records to include only highly rated properties. The GROUP BY location clause groups the filtered records by location, and the COUNT() function calculates the total number of such properties in each location.

This helps identify locations with highly rated listings, which is useful for analyzing customer satisfaction and finding top-performing destinations.

10. Find the average rating of Airbnb properties for each property type.

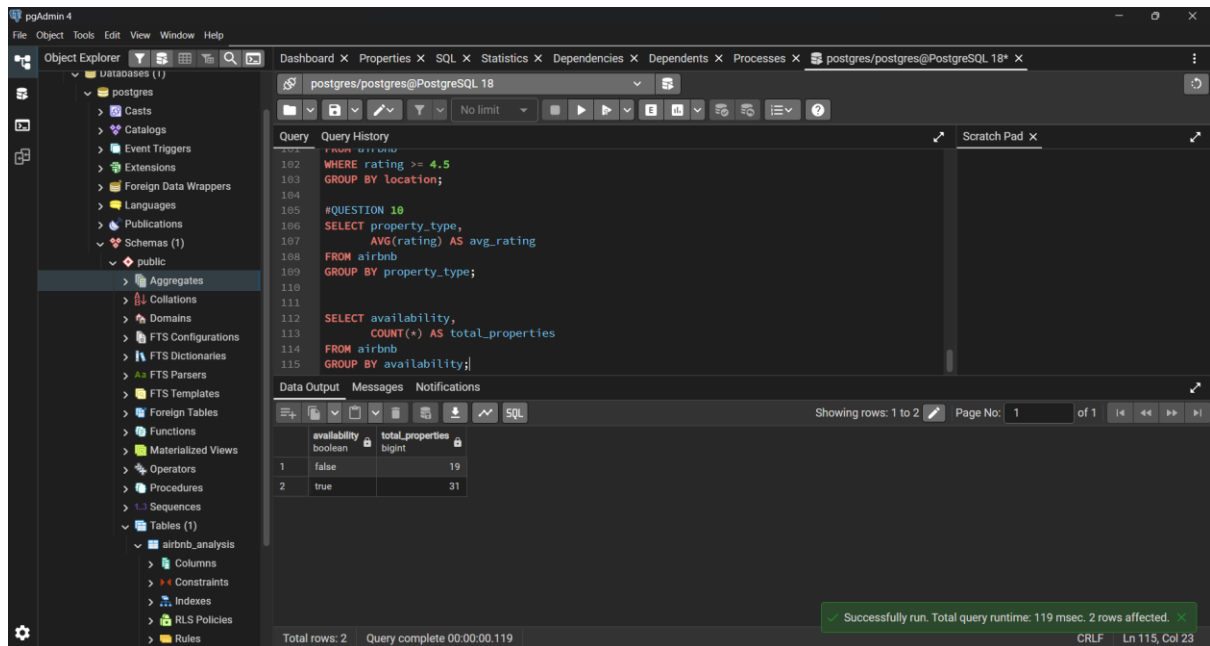


```
select property_name,  
       avg(rating) as avg_rating  
from airbnb  
group by property_name;
```

This query calculates the average customer rating for each type of property using the AVG() function. The GROUP BY property_name clause groups all records based on property type, such as Apartment, Villa, Cottage, etc., and then computes the average rating for each group.

This helps compare customer satisfaction across different types of properties and identify which property types generally receive higher ratings. It is useful for performance analysis and understanding customer preferences.

11. Find the number of available and unavailable properties.

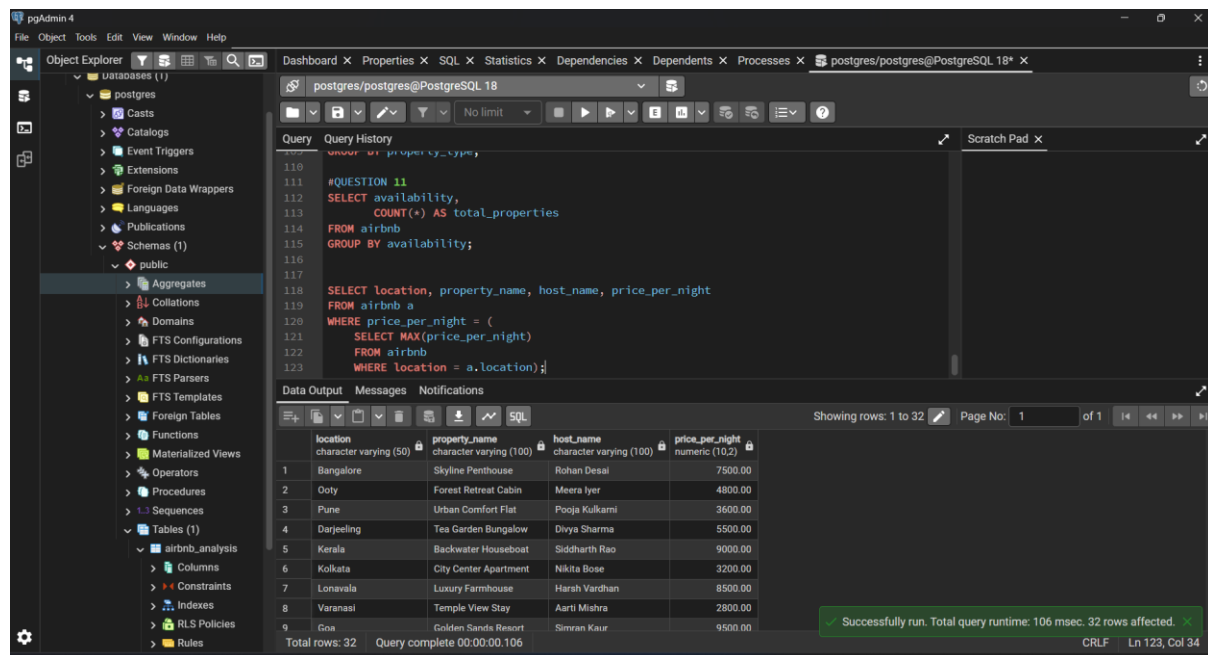


```
select availability,  
count(*) as total_properties  
from airbnb  
group by availability;
```

This query counts the total number of properties based on their availability status using the COUNT() function. The GROUP BY availability clause groups the records into two categories: available (TRUE) and unavailable (FALSE).

This helps understand how many properties are currently available for booking and how many are not, which is useful for analyzing availability trends and overall property status.

12. Find the highest-priced property in each location.

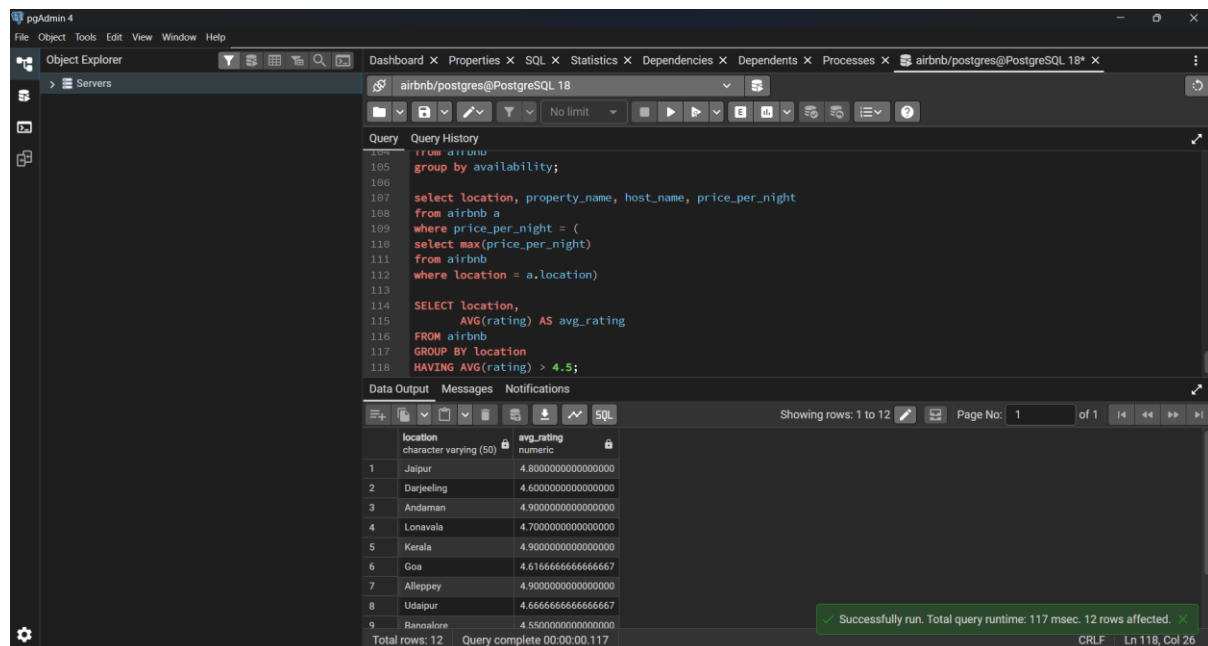


```
select location, property_name, host_name, price_per_night
from airbnb a
where price_per_night = (
select max(price_per_night)
from airbnb
where location = a.location)
```

This query retrieves the most expensive property in each location using a subquery. The inner query `SELECT MAX(price_per_night)` finds the highest price for each specific location. The outer query then displays the property name, host name, location, and price for those highest-priced properties.

This helps identify premium properties in every city, making it useful for market comparison, pricing analysis, and identifying luxury listings in each location.

13. Find locations where the average rating is greater than 4.5.

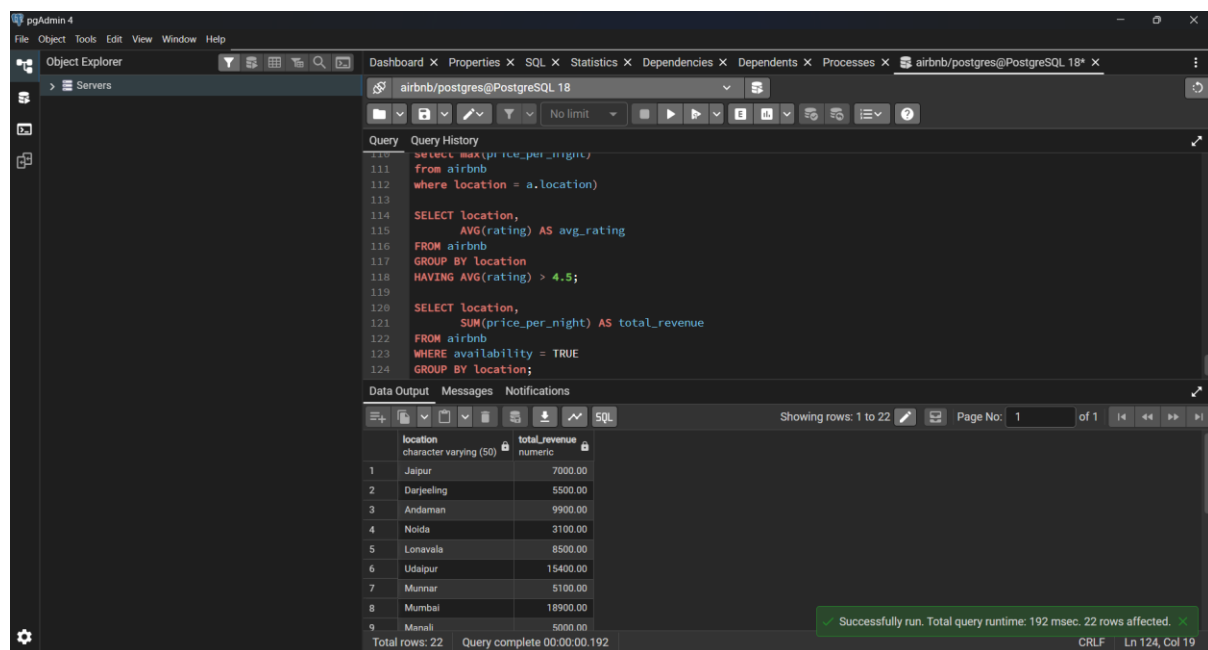


```
SELECT location,
        AVG(rating) AS avg_rating
FROM Airbnb
GROUP BY location
HAVING AVG(rating) > 4.5;
```

This query calculates the average rating for each location using the `AVG(rating)` function. The `GROUP BY location` clause groups records based on each location. The `HAVING AVG(rating) > 4.5` condition filters the grouped results to display only those locations whose average rating exceeds 4.5.

This helps identify highly rated destinations, making it useful for analyzing customer satisfaction and determining the best-performing cities based on overall guest reviews.

14. Find the total revenue potential per location (price × number of available properties).

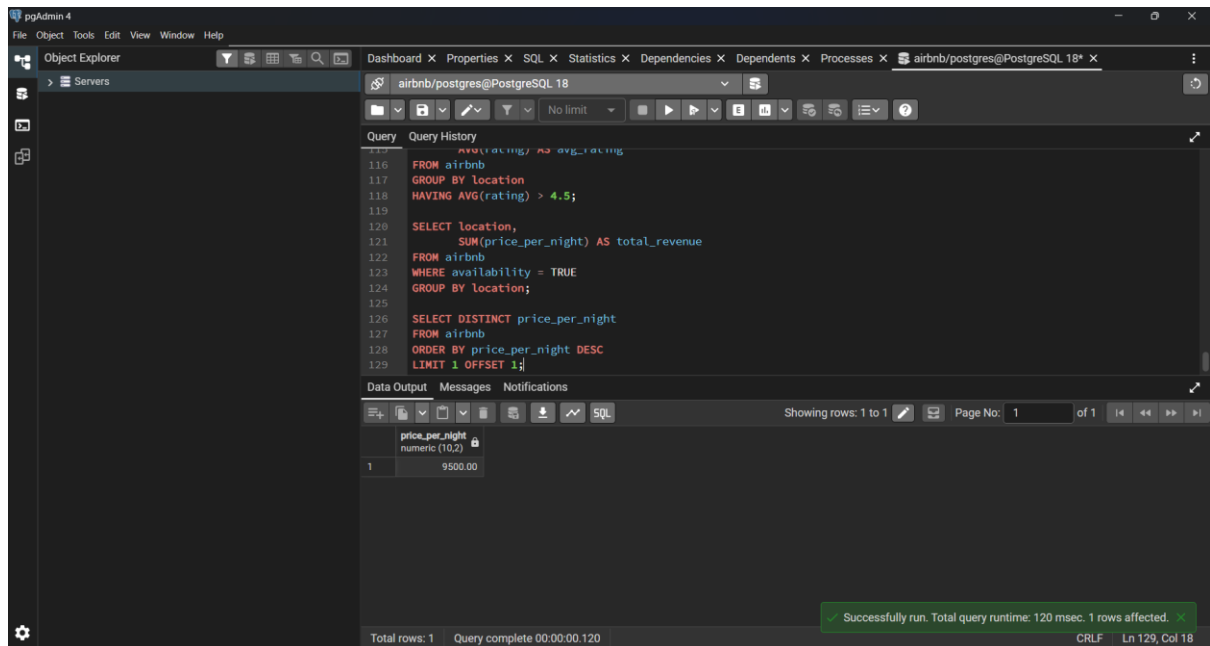


```
SELECT location,
        SUM(price_per_night) AS total_revenue
FROM airbnb
WHERE availability = TRUE
GROUP BY location;
```

This query calculates the total potential revenue for each location by summing the price_per_night of properties that are currently available. The WHERE availability = TRUE clause filters only available properties. The SUM(price_per_night) function adds their prices, and GROUP BY location organizes the results by location.

This helps estimate the possible daily revenue from active listings in each city, which is useful for business analysis and understanding income potential.

15. Find the second highest price property.

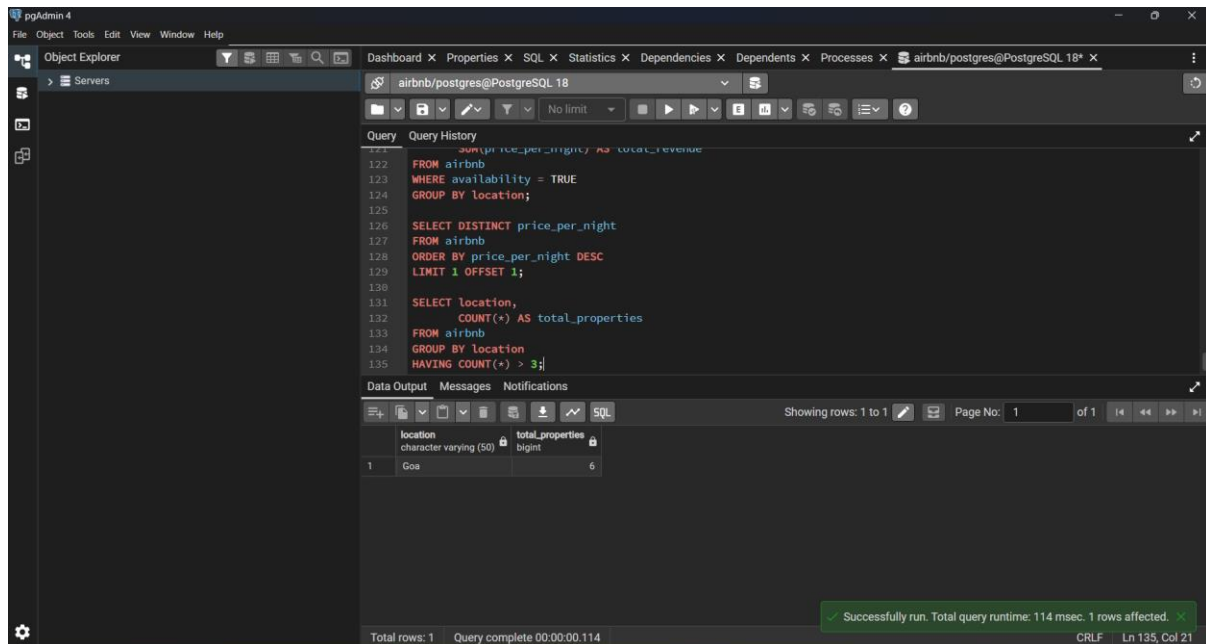


```
SELECT DISTINCT price_per_night  
FROM airbnb  
ORDER BY price_per_night DESC  
LIMIT 1 OFFSET 1;
```

This query retrieves the second highest price in the dataset. The `ORDER BY price_per_night DESC` clause sorts prices from highest to lowest. The `LIMIT 1 OFFSET 1` skips the highest price and returns the next one. The `DISTINCT` keyword ensures duplicate prices are not repeated.

This helps identify properties that are nearly the most expensive, which is useful for ranking analysis and price comparison.

16. Find locations having more than 3 properties.

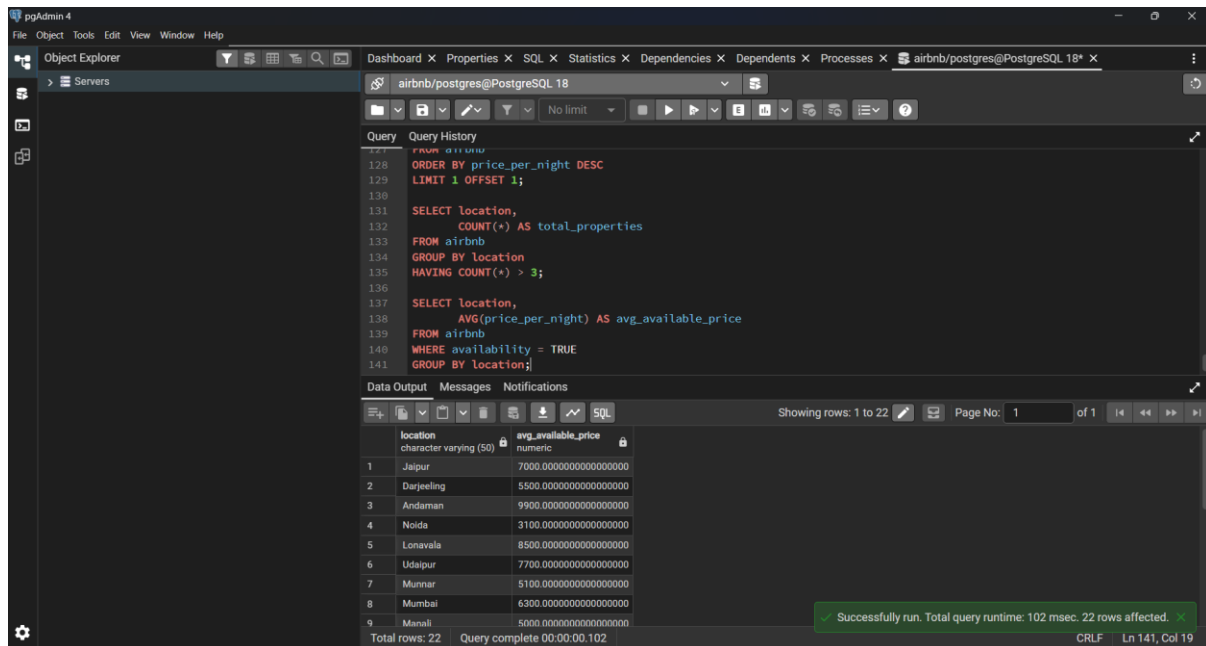


```
SELECT location,  
      COUNT(*) AS total_properties  
FROM airbnb  
GROUP BY location  
HAVING COUNT(*) > 3;
```

This query counts the number of properties in each location using the COUNT(*) function. The GROUP BY location clause groups the data by location. The HAVING COUNT(*) > 3 condition filters only those locations that have more than three properties.

This helps identify cities with a high concentration of listings, which is useful for analyzing market density and determining popular destinations with strong property supply.

17. Find the average price of available properties in each location.

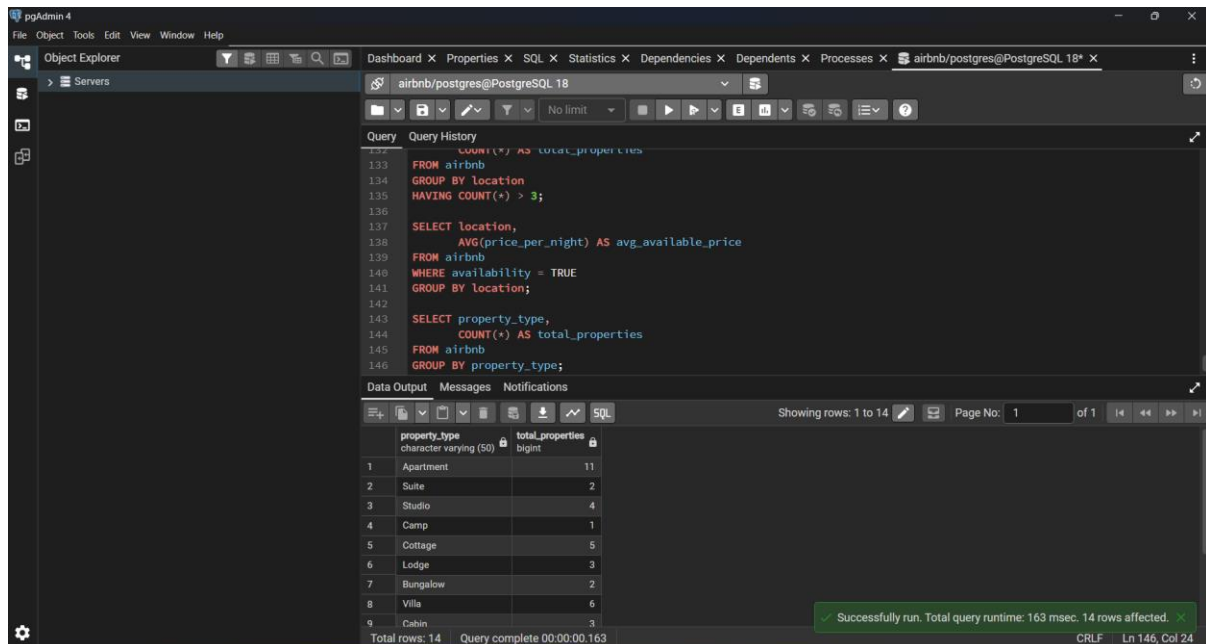


```
SELECT location,  
      AVG(price_per_night) AS avg_available_price  
FROM airbnb  
WHERE availability = TRUE  
GROUP BY location;
```

This query calculates the average price per night for properties that are currently available. The WHERE availability = TRUE clause filters only available listings. The AVG(price_per_night) function computes the average price, and GROUP BY location organizes the results by location.

This helps analyze active market pricing and compare the average cost of available properties across different cities.

18. Find the total number of properties for each property type.

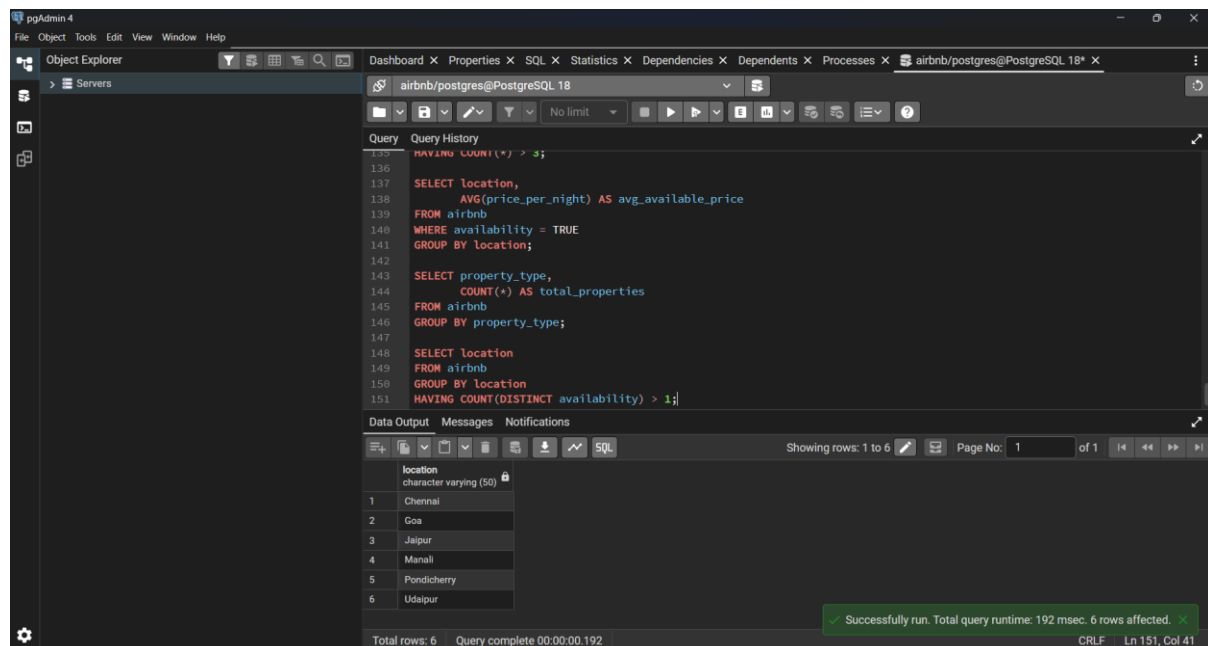


```
SELECT property_type,
       COUNT(*) AS total_properties
FROM airbnb
GROUP BY property_type;
```

This query counts how many properties exist under each property type using the COUNT() function. The GROUP BY property_type clause groups listings based on their type such as Apartment, Villa, Cottage, etc.

This helps understand which types of properties are most common in the dataset and analyze market trends based on property categories.

19. Find locations where both available and unavailable properties exist.

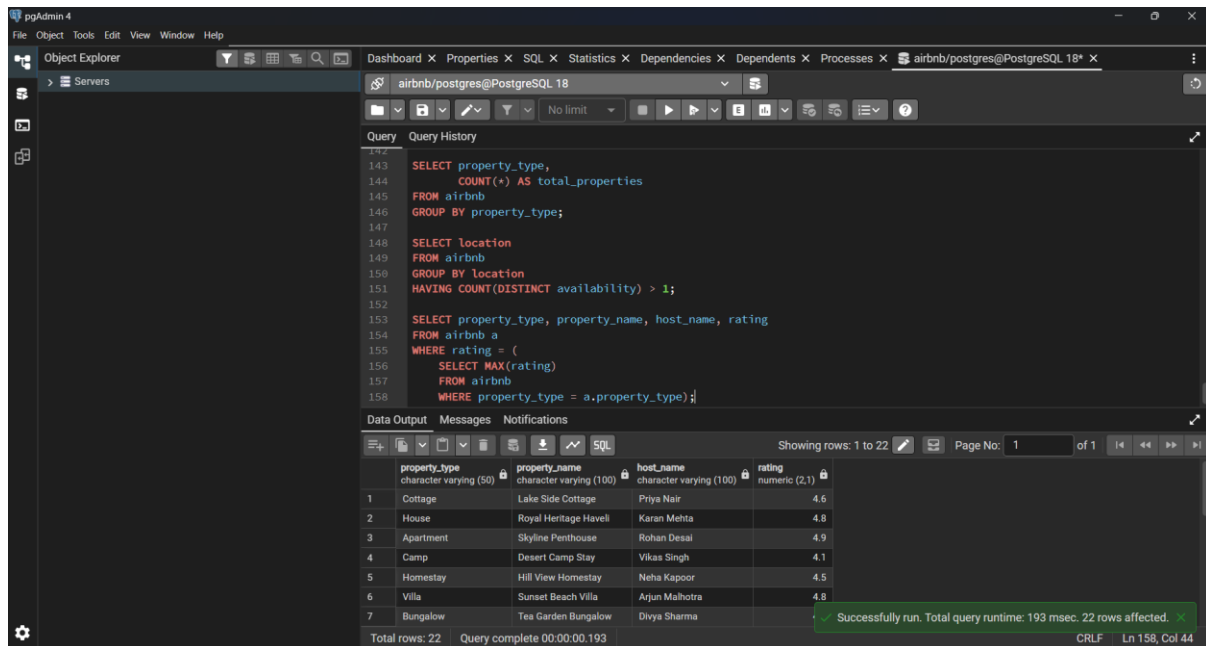


```
SELECT location  
FROM airbnb  
GROUP BY location  
HAVING COUNT(DISTINCT availability) > 1;
```

This query groups records by location and checks whether both availability statuses (TRUE and FALSE) are present. The `COUNT(DISTINCT availability)` counts different availability values in each location. If the count is greater than 1, it means both available and unavailable properties exist there.

This helps identify balanced markets where some properties are booked while others remain available.

20. Find the top-rated property in each property type.



The screenshot shows the pgAdmin 4 interface with a SQL query executed against the 'airbnb/postgres@PostgreSQL 18' database. The query is as follows:

```
143 SELECT property_type,  
144        COUNT(*) AS total_properties  
145 FROM airbnb  
146 GROUP BY property_type;  
147  
148 SELECT location  
149 FROM airbnb  
150 GROUP BY location  
151 HAVING COUNT(DISTINCT availability) > 1;  
152  
153 SELECT property_type, property_name, host_name, rating  
154 FROM airbnb a  
155 WHERE rating = (  
156     SELECT MAX(rating)  
157     FROM airbnb  
158     WHERE property_type = a.property_type);
```

The 'Data Output' pane shows the results of the query, displaying 22 rows. The columns are: property_type (character varying (50)), property_name (character varying (100)), host_name (character varying (100)), and rating (numeric (2,1)).

	property_type	property_name	host_name	rating
1	Cottage	Lake Side Cottage	Priya Nair	4.6
2	House	Royal Heritage Haveli	Karan Mehta	4.8
3	Apartment	Skyline Penthouse	Rohan Desai	4.9
4	Camp	Desert Camp Stay	Vikas Singh	4.1
5	Homestay	Hill View Homestay	Neha Kapoor	4.5
6	Villa	Sunset Beach Villa	Arjun Malhotra	4.8
7	Bungalow	Tea Garden Bungalow	Divya Sharma	

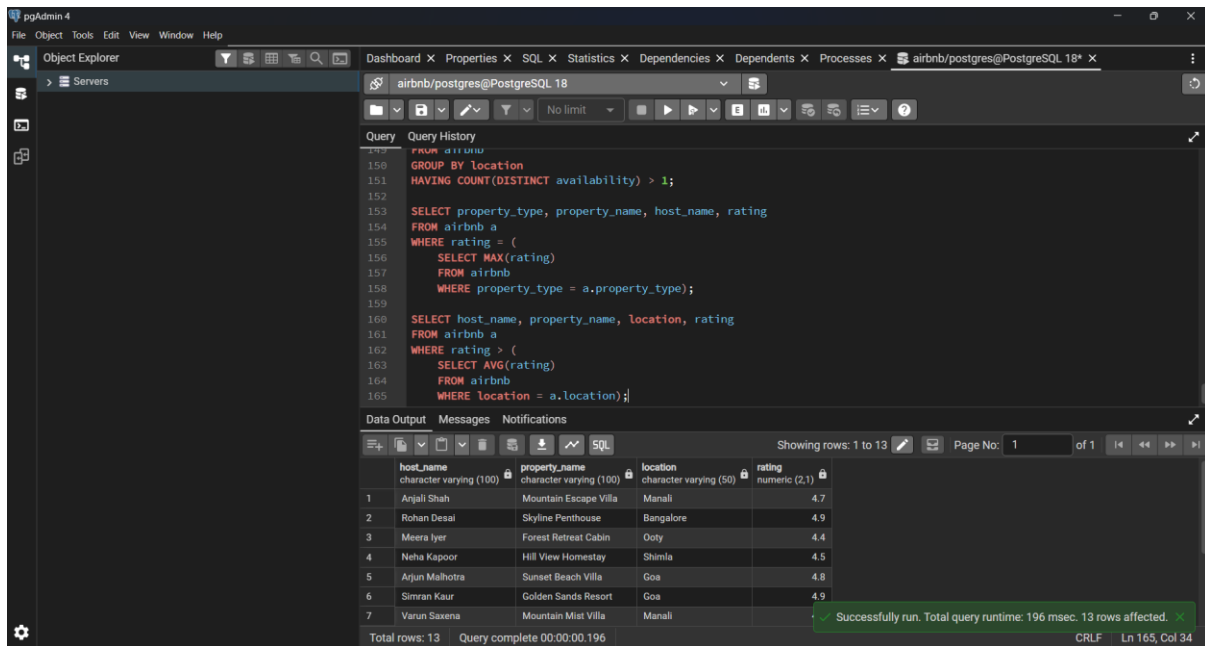
At the bottom, a status bar indicates: 'Total rows: 22', 'Query complete 00:00:00.193', and 'Successfully run. Total query runtime: 193 msec. 22 rows affected.'

```
SELECT property_type, property_name, host_name, rating  
FROM airbnb a  
WHERE rating = (  
    SELECT MAX(rating)  
    FROM airbnb  
    WHERE property_type = a.property_type);
```

This query retrieves the highest-rated property within each property type using a subquery. The inner query finds the maximum rating for each property type. The outer query then displays details of properties that match that highest rating.

This helps identify the best-performing property in each category, useful for quality comparison and customer preference analysis.

21. Find properties whose rating is above the average rating of their respective location.



```
SELECT host_name, property_name, location, rating
FROM airbnb a
WHERE rating > (
  SELECT AVG(rating)
  FROM airbnb
  WHERE location = a.location);
```

This query compares each property's rating with the average rating of its location. The inner subquery calculates the average rating for each location, and the outer query selects properties with ratings higher than that average.

This helps identify above-average performing properties within each city, useful for competitive analysis and identifying high-quality listings.

CHAPTER 4: KEY FINDINGS

The study revealed several important findings regarding the performance and application of PostgreSQL in managing and analyzing Airbnb rental data:

- **Performance:** PostgreSQL efficiently handled data storage, retrieval, and analysis of Airbnb listings. Queries such as finding average price, highest-priced properties, and top-rated listings were executed accurately and efficiently, demonstrating PostgreSQL's strong performance in handling structured business data.
- **Data Management Efficiency:** PostgreSQL provided an effective platform for organizing Airbnb rental data into structured tables. It allowed easy insertion, updating, filtering, and retrieval of data based on attributes such as location, price, property type, and rating, ensuring efficient database management.
- **Data Integrity and Reliability:** PostgreSQL ensured data consistency and accuracy through its ACID compliance and constraint support. This helped maintain reliable and error-free records of property listings, host details, pricing, and availability, which is essential for business decision-making.
- **Analytical Capability:** PostgreSQL enabled meaningful data analysis using SQL queries such as AVG(), COUNT(), GROUP BY, ORDER BY, and filtering conditions. These queries helped identify pricing trends, top-performing properties, property distribution, and availability patterns across different locations.
- **Scalability and Real-World Application:** PostgreSQL demonstrated the capability to handle rental management systems effectively. It can be used in real-world applications such as Airbnb-like platforms, hotel management systems, and online rental marketplaces where large volumes of structured data need to be stored and analyzed.
- **Decision-Making Support:** The analysis performed using PostgreSQL provided valuable insights such as identifying premium properties, highly rated listings, and location-based pricing trends. These insights can help business owners, customers, and managers make informed decisions.

CHAPTER 5: CONCLUSION

This study successfully demonstrated the use of PostgreSQL in designing and managing an Airbnb rental database system. PostgreSQL proved to be a reliable and efficient database management system for storing, organizing, and analyzing structured data related to property listings, hosts, pricing, ratings, and availability. Through the creation of tables and execution of various SQL queries, it was possible to retrieve meaningful information such as average prices, highest-priced properties, top-rated listings, and property distribution across different locations.

The project also showed that PostgreSQL ensures data accuracy, consistency, and integrity, which are essential for real-world business applications. It helped in analyzing rental data effectively and provided useful insights that can support better decision-making for customers and business owners. Overall, PostgreSQL is a powerful and suitable tool for managing rental management systems and can be effectively used in real-world applications such as Airbnb, hotel booking systems, and other online platforms that require efficient database management and analysis.

CHAPTER 6: BIBLIOGRAPHY

- Cheng, M., & Foley, C. (2019). Algorithmic management: The case of Airbnb. *International Journal of Hospitality Management*.
- Deboosere, R., Kerrigan, D. J., Wachsmuth, D., & El-Geneidy, A. (2019). Location, location and professionalization: A multilevel hedonic analysis of Airbnb listing prices and revenue. *Regional Studies, Regional Science*, 6(1), 143–156.
- Dolnicar, S. (2018). *Peer-to-peer accommodation networks: Pushing the boundaries*. Goodfellow Publishers. <https://doi.org/10.23912/9781911396512>
- Gong, S. (2023). *Technology adoption in the short-term rental property management industry* (Master's dissertation). Universidade de Lisboa.
- Horton, T. (2016). *Reducing affordability: The impact of Airbnb on the vacancy rate and affordability of the Toronto rental market* (Major paper). University of Ottawa.
- Hübscher, M., & Feiler, A. (2025). A fair Airbnb? (Self-)perceptions in the Fairbnb cooperative that aims to revolutionise the short-term rental market. *Current Issues in Tourism*.
- Leoni, G., & Parker, L. D. (2019). Governance and control of sharing economy platforms: Hosting on Airbnb. *The British Accounting Review*, 51(6), 100814. <https://doi.org/10.1016/j.bar.2018.12.001>
- Li, J., & Biljecki, F. (2019). The implementation of big data analysis in regulating online short-term rental business: A case of Airbnb in Beijing. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*
- Mody, M., & Gomez, M. (2018). Airbnb and the hotel industry: The past, present, and future of sales, marketing, branding, and revenue management. *Boston Hospitality Review*. Boston University.
- Ndaguba, E. A., & Van Zyl, C. (2024). Data-driven approach to optimizing property management strategies: Spatial modeling analytics of short-term rentals. *International Journal of Housing Markets and Analysis*. <https://doi.org/10.1108/IJHMA-08-2024-0117>
- Obrč, P., & Kerbler, B. (2021). The impact of Airbnb on long-term rental housing: The case of Ljubljana. *Housing Critical*, 8(1), 150–158.
- Sainaghi, R., & d'Angella, F. (Year). *Marketing capabilities for value creation in the short-term rental segment: An exploratory analysis of Airbnb*.

Sims, J., Ameen, N., & Bauer, R. (Year). *Dynamic pricing and benchmarking in Airbnb*.

Śpiewak-Szyjka, M., & Oleńczuk-Paszal, A. (2024). The specifics of short-term rental property management. *Scientific Papers of Silesian University of Technology, Organization and Management Series*, No. 209.

Swor, R. W. F. (2019). Long-term solutions to the short-term problem: An analysis of the current legal issues related to Airbnb and similar short-term rental companies with a proposed model ordinance. *Belmont Law Review*.

Wang, Y., Livingston, M., McArthur, D. P., & Bailey, N. (2024). The challenges of measuring the short-term rental market: An analysis of open data on Airbnb activity. *Housing Studies*, 39(9), 2260–2279.

Filieri, R (2023). The impact of COVID-19 on Airbnb demand and pricing dynamics: Evidence from large-scale booking data. *Annals of Tourism Research*, 98, 103523.